

Element D

May 15 - edited foldability of the board to now just be flat

Mancala game - Made of wood, paint, and epoxy resin. and requires only marbles along with the board

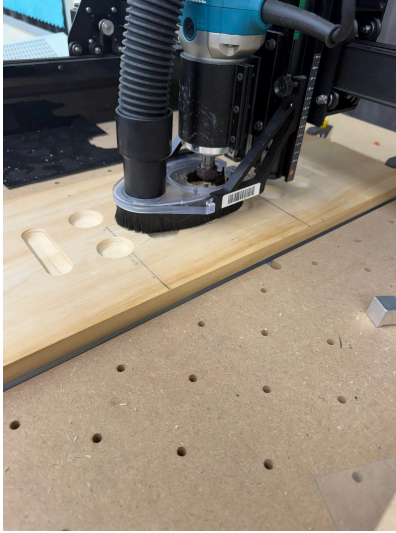
Simple game, fun physical game, easy to have in a school due to size

Dimensions: 18 x 4.5 in

Materials:

- Wood
- CNC Machine
- Hand saw
- 48 marbles/beads
- Yard stick
- Pen
- Epoxy resin (purple + gold)
- Black paint
- Sand paper

Documentation of Progression:







Primary Requirements:

Playability

Proportional

Sturdy

When deciding on our idea, we needed to make sure it could well fit the primary requirements of the project. Mancala came to mind as it is a game that does not take up a lot of space and that in itself is a very physically sturdy game. The board is wooden and has to be thick to allow us to make pits in it. It is also a universally understandable game, making it playable as long as instructions are still provided. It is also proportional as we create it using specific measurements and sketches to make sure any cutting and shaping is lined up well. We also

were prepared to sand this board knowing it would benefit its proportions and playability if it is smoothened.

We will make an online 3D design of our game.

Then, we upload a file to cut out the divots of the game into a piece of wood measured 18 x 4.5 inches.

After that, we will use a hand saw to cut out the board

We will sand the board and smooth the corners and pits

We will paint the board and fill the pits with a thin layer of resin

We will create, print, and laminate the instructions

CAD Model:

